MPTCP vs MP-QUIC: A Comparative Study of Multipath Transport Protocols

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Kaushal Rathi Roll No. 24110290	Krina Paghadar Roll No. 24110169	Pratik Kumar Dhakar Roll No. 24110277	Swamini More Roll No. 24110211
QUIC & MP-QUIC Design; Comparative Analysis Lead	*MPTCP Architecture, Linux Kernel Status & Adoption Lead*	*Coupled Congestion Control, Scheduling & Methodology Lead*	*Published Research Synthesis & Application Implications Lead*

[NOTE: The report is strictly limited to a maximum of 10 pages. All sections outlined in this template are mandatory and must be included in your report. The descriptions provided under each section are key suggestions. You may adapt and draft the specific content to best fit your project's requirements. All text must be written in Times New Roman/Arial, size 10.]

I. Introduction

Modern user equipment (UE) and edge computing platforms—including smartphones, connected vehicles, and portable workstations—are overwhelmingly multihomed, concurrently equipped with multiple heterogeneous physical network interfaces such as Wi-Fi, 5G/LTE cellular, and wired Ethernet. Similarly, hyperscale cloud data centers leverage multi-rooted Clos and fat-tree topologies that offer dense arrays of parallel, disjoint physical paths between server racks. Despite the physical presence of multiple concurrent paths, traditional transport-layer protocols remain fundamentally single-path: Transmission Control Protocol (TCP, RFC 793 / RFC 9293) and standard QUIC (RFC 9000) bind a logical connection immutably to a single network path defined by an IP 4-tuple (source IP, source port, destination IP, destination port). Consequently, when a mobile terminal moves out of Wi-Fi range, active transport connections stall or abruptly terminate, forcing costly application-level reconnects while leaving alternate physical capacity entirely idle ***[Specification: RFC 6182, Section 1]***.

To overcome these architectural constraints, two prominent transport-layer multipath protocols have emerged: Multipath TCP (MPTCP v1, RFC 8684) and Multipath QUIC (MP-QUIC, currently specified in draft-ietf-quic-multipath-21). MPTCP was designed to extend TCP incrementally by multiplexing multiple underlying TCP subflows beneath a single socket abstraction, preserving backwards compatibility with the massive ecosystem of legacy socket applications. In contrast, MP-QUIC represents a clean-slate, user-space transport protocol layered over UDP, incorporating native cryptographic encapsulation via TLS 1.3 and introducing explicit multi-path management at the frame level ***[Specification: draft-ietf-quic-multipath-21, Section 1]***.

Despite more than a decade of research and standards development, multipath transport protocols remain rarely utilized in open Internet applications. Enterprise middleboxes frequently strip unknown TCP options, mobile operating systems restrict multipath APIs to proprietary first-party daemons (such as Apple Siri), and hostile network firewalls routinely drop or rate-limit UDP traffic ***[Published Research: IMC 2011; arXiv:2106.07351]***. Furthermore, transmitting across paths with divergent round-trip times (RTTs) introduces severe Head-of-Line (HOL) blocking and receive-window collapse, frequently causing multi-path transfers to perform worse than the best single path ***[Published Research: CoNEXT 2017]***.

This study provides a rigorous, first-principles comparative evaluation of MPTCP v1 and MP-QUIC. We deconstruct their connection establishment mechanics, sequence number spaces, loss recovery procedures, coupled congestion control dynamics, and operational deployability. Through a meticulous synthesis of published peer-reviewed empirical studies and official IETF standards, we identify the exact network conditions under which each protocol succeeds or fails, providing concrete transport-selection recommendations for bulk transfer, adaptive video streaming, and real-time voice communications.

II. Problem Statement

The overarching objective of this project is to conduct an in-depth, defensible comparative evaluation between Multipath TCP (MPTCP v1, RFC 8684) and Multipath QUIC (draft-ietf-quic-multipath-21) across three fundamental dimensions: architectural design, real-world deployability across hostile Internet paths, and protocol resiliency under path heterogeneity (asymmetric RTT, packet loss, and dynamic link failure) ***[Our Analysis]***.

Scope and Methodological Boundaries (Strictly Literature and Theory): This investigation is strictly a theoretical, architectural, and published literature study. The student team has authored no transport protocol code, configured no operational multipath operating system kernels, and conducted no live physical or emulated testbed experiments. Initial attempts to initialize an empirical Linux testbed on host machines encountered insuperable platform blockers: the stock Linux kernel provided by Windows Subsystem for Linux (WSL2) omits MPTCP compilation flags (CONFIG_MPTCP=y is absent), and an alternative Canonical Multipass Ubuntu virtual machine encountered fatal hypervisor initialization failures on Windows. Rather than investing project time troubleshooting virtualization drivers or generating unreproducible toy data, the team strictly re-scoped the work into an exhaustive, first-principles deconstruction of primary RFCs, IETF drafts, and peer-reviewed literature ***[Our Analysis]***.

Inputs, Constraints, and Scope Definition: The analytical inputs comprise primary standards track specifications—RFC 8684 (MPTCP v1), RFC 6824 (MPTCP v0), RFC 6356 (Coupled Congestion Control / LIA), RFC 9000/9002 (QUIC), and draft-ietf-quic-multipath-21—alongside peer-reviewed empirical papers (CoNEXT 2017, MultipathTester 2019, IMC 2011). The outputs comprise formal comparative architectural matrices, mathematical fairness analyses, empirical synthesis tables, and defensible application recommendations. The operational constraints analyzed include backwards compatibility, the strict 40-byte TCP option budget, middlebox interference, UDP filtering, and user-space vs. kernel-space CPU processing costs ***[Our Analysis]***.

Gap in Existing Literature: A substantial portion of published academic literature evaluates obsolete protocol variants—specifically MPTCP v0 (RFC 6824) implemented in the legacy out-of-tree Linux kernel (v0.96) or pre-IETF Google QUIC forks (CoNEXT 2017). These historic evaluations cannot be directly extrapolated to today's upstream Linux kernel (v5.6+, RFC 8684) or the mature IETF MP-QUIC working group draft (draft-21). This study explicitly bridges this gap by reconciling historical empirical findings with current, active standards specifications ***[Our Analysis]***.

III. Project Objectives

To ensure rigorous, checkable academic execution, this study defines five specific analytical objectives ***[Our Analysis]***:

1. Architectural Deconstruction: Formally map and compare the connection lifecycle, handshake state machines, address signaling, and teardown mechanisms of MPTCP v1 (RFC 8684) and MP-QUIC (draft-ietf-quic-multipath-21).

2. Sequence Mapping and Loss Detection Analysis: Analytically deconstruct MPTCP's two-level sequence space (SSN to DSN via DSS) versus MP-QUIC's dedicated per-path Packet Number (PN) spaces and PATH_ACK frames, establishing how each architecture handles retransmission ambiguity and Head-of-Line blocking under delay asymmetry.

3. Congestion Control and Fairness Evaluation: Mathematically analyze Coupled Congestion Control algorithms (RFC 6356 LIA, OLIA, BALIA) to evaluate Pareto-optimality, 'do-no-harm' fairness to single-path flows at shared bottlenecks, and resource pooling dynamics.

4. Deployability and Middlebox Assessment: Characterize real-world Internet deployability barriers, evaluating middlebox TCP option stripping, sequence mangling, NAT rebinding, UDP firewall blocking, and operating system integration overhead.

5. Synthesis of Published Empirical Evidence: Systematically synthesize published experimental evaluations from peer-reviewed literature (CoNEXT 2017, MultipathTester 2019, Paasch 2014) across diverse traffic workloads (bulk file transfer, DASH/HLS adaptive video, and real-time VoIP), identifying conflicting results and boundaries of validity.

V. Architecture and Design

A. Multipath TCP Architecture (RFC 8684):

MPTCP v1 is designed as an in-kernel transport extension that presents a standard, single-socket byte-stream interface (SOCK_STREAM) to application programs while multiplexing data transmission across multiple underlying TCP subflows ***[Specification: RFC 8684, Section 1.1]***. Each subflow operates as a standard TCP connection with its own independent IP 4-tuple, congestion window (cwnd), round-trip time estimator (SRTT), and retransmission timer (RTO).

To reconcile subflow-level TCP semantics with connection-wide ordered delivery, MPTCP implements a dual-level sequence numbering architecture ***[Specification: RFC 8684, Section 3.3]***:

1. Subflow Sequence Number (SSN, 32-bit): Governs standard TCP hop-by-hop reliability and flow control over each individual subflow. Middleboxes observe standard TCP sequence numbers, preventing firewall resets.

2. Data Sequence Number (DSN, 64-bit): Governs connection-level reassembly across all subflows, ensuring the receiving application receives an unbroken, in-order byte stream.

The translation between subflow bytes and connection-level data is achieved through the Data Sequence Signal (DSS) option ***[Specification: RFC 8684, Section 3.3]***. The DSS carries explicit 4-tuples: Data Sequence Number, Subflow Sequence Number, Data-Level Length, and a 64-bit Data ACK. Because middleboxes frequently split or coalesce TCP segments, DSS mappings define byte intervals rather than per-packet associations. Furthermore, MPTCP v1 supports an optional DSS checksum to detect middlebox payload rewriting. If an on-path proxy alters payload data without updating the DSS, the checksum fails, triggering an MP_FAIL option and a permanent fallback to plain TCP using an infinite mapping ***[Specification: RFC 8684, Section 3.6]***.

Connection establishment in MPTCP v1 occurs via the MP_CAPABLE option embedded in the SYN, SYN/ACK, and ACK packets of the initial subflow ***[Specification: RFC 8684, Section 3.1]***. Each endpoint generates a 64-bit cryptographic key. Unlike MPTCP v0 (RFC 6824), MPTCP v1 mandates that the initiator echo both keys in the final ACK of the three-way handshake, enabling stateless server operation (SYN cookies) without pre-allocating connection state. Subsequent subflows are joined via the MP_JOIN option, which authenticates the join request using an HMAC over a shared secret derived from the receiver's key token (a truncated 32-bit SHA-256 hash) and random nonces, preventing off-path session hijacking ***[Specification: RFC 8684, Section 3.2]***. Additional path management is achieved via ADD_ADDR and REMOVE_ADDR options, which are authenticated using HMACs to defeat spoofed routing attacks ***[Specification: RFC 8684, Section 3.4]***.

B. Multipath QUIC Architecture (draft-ietf-quic-multipath-21):

In stark contrast to MPTCP, MP-QUIC operates entirely in user space layered over UDP, encapsulating all transport metadata and stream frames inside TLS 1.3 authenticated and encrypted packets ***[Specification: RFC 9000; draft-ietf-quic-multipath-21, Section 1]***. In MP-QUIC, a 'path' is formally defined as an end-to-end communication channel identified by a 32-bit Path ID and governed by a distinct Connection ID (CID) ***[Specification: draft-ietf-quic-multipath-21, Section 3]***. Multipath capability is negotiated during the initial TLS handshake via the initial_max_path_id transport parameter; exchanging non-zero values enables multipath operation.

The defining architectural breakthrough of draft-ietf-quic-multipath-21 is the adoption of dedicated, independent Packet Number (PN) spaces per path ***[Specification: draft-ietf-quic-multipath-21, Section 4]***. In earlier research designs (such as the 2017 CoNEXT prototype), a single connection-wide PN space was shared across all paths, which caused severe distortion in RTT estimation, loss detection, and congestion window updates when packets arrived out of order across asymmetric paths. Under draft-21, each path maintains its own monotonically increasing PN space starting at 0. Because QUIC never reuses a packet number upon retransmission (transmitting lost stream data in a new packet with a new PN and original stream offset), retransmission ambiguity is completely eliminated ***[Specification: RFC 9002; draft-ietf-quic-multipath-21, Section 6]***.

MP-QUIC introduces specialized control frames for multipath lifecycle management ***[Specification: draft-ietf-quic-multipath-21, Section 5]***:

1. PATH_ACK: Carries acknowledgment ranges for a specific path ID. Crucially, draft-21 permits PATH_ACK frames to be transmitted across any active path, enabling low-latency ACK delivery via a fast path for packets sent over a high-latency path.

2. PATH_STATUS: Explicitly signals whether a path is in AVAILABLE or BACKUP status, containing a sequence number to prevent out-of-order state transitions.

3. PATH_ABANDON: Instructs the peer to immediately abandon an impaired path, specifying an error code. Endpoints retain path context for three Probe Timeouts (3 * PTO) to absorb trailing delayed packets.

4. PATH_CHALLENGE / PATH_RESPONSE: Validates path reachability and enforces the anti-amplification limit, ensuring an endpoint sends at most three times the validated byte volume prior to address verification ***[Specification: RFC 9000, Section 8.1]***.

C. Architectural Comparison and Trade-offs:

The contrasting design philosophies between MPTCP v1 and MP-QUIC manifest in distinct architectural trade-offs across backwards compatibility, middlebox robustness, and stream multiplexing ***[Our Analysis]***:

Table 1: Architectural Comparison — MPTCP v1 (RFC 8684) vs. MP-QUIC (draft-ietf-quic-multipath-21)

Architectural Dimension	Multipath TCP (MPTCP v1)	Multipath QUIC (MP-QUIC)	Primary Citation & Section
Protocol Layering	In-kernel transport layer; extension to TCP over IPv4/IPv6	User-space transport library layered over UDP; encrypted via TLS 1.3	RFC 8684 §1.1; draft-21 §1
Application API	Standard POSIX socket (SOCK_STREAM); transparent to legacy applications	Stream-based user-space API; requires application recompilation / integration	RFC 8684 §3.1; RFC 9000 §2
Path Identification	Implicitly bound to TCP subflow IP 4-tuples; signaled via ADD_ADDR	Explicit 32-bit Path ID; distinct Connection IDs (CIDs) per path	RFC 8684 §3.4; draft-21 §3
Connection Setup	3-way handshake over primary subflow via MP_CAPABLE option (1 RTT)	Combined transport & TLS 1.3 handshake via CRYPTO frames (1 RTT / 0-RTT)	RFC 8684 §3.1; draft-21 §3
Adding Subflows / Paths	MP_JOIN option on new 4-tuple; authenticated via HMAC and receiver token	PATH_CHALLENGE / RESPONSE; validated via 4-way ping-pong & anti-amplification	RFC 8684 §3.2; draft-21 §5.1
Sequence Number Spaces	Dual-level: 32-bit Subflow Sequence (SSN) + 64-bit Data Sequence (DSN)	Independent Packet Number (PN) space per path + per-stream byte offsets	RFC 8684 §3.3; draft-21 §4
Sequence Mapping	Explicit DSS (Data Sequence Signal) option mapping SSN intervals to DSN	Implicit mapping: STREAM frames carry Stream ID and Stream Offset	RFC 8684 §3.3; RFC 9000 §19.8
Loss Detection & ACKs	Per-subflow TCP duplicate ACKs / SACK + connection-level Data ACK	Dedicated PATH_ACK frames per path; ACKs may return over alternate paths	RFC 8684 §3.3; draft-21 §5.2
Retransmission Ambiguity	Present at subflow level; requires TCP timestamps option to resolve	Completely eliminated; packet numbers never repeat upon retransmission	RFC 6298; draft-21 §6
HOL Blocking Scope	Connection-wide: packet loss on any subflow blocks all subflow delivery	Stream-isolated: loss on one stream does not block concurrent streams	CoNEXT 2017 §4; draft-21 §1
Middlebox Vulnerability	High: firewalls strip unknown TCP options; proxies mangle sequence numbers	Immune to option stripping; payload & headers encrypted; UDP drop risk	IMC 2011; draft-21 §1
Congestion Control	Coupled Congestion Control (RFC 6356 LIA, OLIA, BALIA) across subflows	Independent per-path congestion control (RFC 9002); coupled algorithms optional	RFC 6356 §3; draft-21 §6.2

VI. Implementation

Status of Protocol Implementations (The Team Authored No Code):

This project analyzes existing production and experimental software implementations. The team authored no protocol software or kernel modifications ***[Our Analysis]***.

A. Multipath TCP Implementations:

1. Upstream Linux Kernel (v5.6+): Multipath TCP was officially merged into the mainline Linux kernel tree in version 5.6 (released March 2020) and is maintained under net/mptcp ***[Existing Implementation: Linux Kernel 5.6+]***. The upstream kernel speaks MPTCP v1 (RFC 8684) exclusively. It decouples socket management from path control: the in-kernel path manager handles basic endpoint addition, while the user-space daemon (mptcpd) manages complex routing policies via generic netlink interfaces. Applications leverage MPTCP transparently via IPPROTO_MPTCP socket options or dynamic library interception (mptcpize).

2. Out-of-Tree Linux Kernel Fork (v0.96): Prior to kernel 5.6, MPTCP was developed as an out-of-tree patch set maintained by UCLouvain (culminating in release v0.96 for kernel 5.4) ***[Existing Implementation: MPTCP v0.96]***. Crucially, this legacy fork implemented MPTCP v0 (RFC 6824). Because the vast majority of historical academic literature (2011–2019) evaluated this legacy fork, their reported throughput figures, sysctl tunables, and scheduler behaviors cannot be directly compared to modern upstream Linux kernels.

3. Commercial Deployments: Apple iOS and macOS represent the largest commercial deployment of MPTCP. Implemented since iOS 7 (2013), Apple restricts MPTCP to specific system daemons—primarily Siri, Apple Maps, and Apple Music—operating strictly in backup/failover mode over cellular interfaces to prevent user data plan exhaustion

[Existing Implementation: Apple iOS / RFC 8041]. In telecommunications, 3GPP ATSSS (Access Traffic Steering, Switching, and Splitting) specifies MPTCP proxy architectures (RFC 8803) for fixed-mobile network convergence.

B. Multipath QUIC Implementations:

1. IETF Standardization Track: MP-QUIC is currently specified in draft-ietf-quic-multipath-21 (updated March 2026) within the IETF QUIC Working Group ***[Specification: draft-ietf-quic-multipath-21]***.

2. Existing Codebases: The original 2017 prototype developed by De Coninck and Bonaventure was implemented as an experimental fork of quic-go over pre-IETF Google QUIC; this codebase is obsolete and unmaintained ***[Existing Implementation: mp-quic 2017 fork]***. In the modern ecosystem, picoquic (developed by Christian Huitema) provides the most comprehensive and actively maintained implementation tracking draft-ietf-quic-multipath ***[Existing Implementation: picoquic]***. Mainline quic-go does not currently support multipath. Experimental prototypes have been demonstrated in Cloudflare's quiche and academic forks, but production browser and operating system deployment remains stalled awaiting finalized RFC publication ***[Our Analysis]***.

VII. Testbed Setup

Methodology of Published Studies Reviewed (The Team Ran No Testbed):

The team conducted no experimental testbed runs. To evaluate protocol behaviors defensibly, we systematically analyzed the experimental methodologies, topologies, and impairment configurations documented in published literature ***[Our Analysis]***.

A. De Coninck & Bonaventure (CoNEXT 2017) Testbed Setup:

The seminal 2017 MPQUIC evaluation utilized an emulated network environment instantiated via Mininet running over a Linux host ***[Published Research: CoNEXT 2017, Section 4]***. The test topology deployed a dual-homed client connected to a dual-homed server via two isolated virtual switch paths. Link characteristics were controlled using Linux traffic control (tc) and NetEm (tc-netem):

1. Bandwidth and Delay Configurations: Path 1 was configured as a baseline low-latency link (10–20 ms RTT, 10–50 Mbps), while Path 2 modeled diverse wireless scenarios including symmetric links (20 ms RTT) and severe delay asymmetry (100 ms RTT modeling LTE/satellite).

2. Loss Profiles: Independent uniform random packet loss rates ranging from 0% to 5% were injected onto individual paths to evaluate loss recovery resilience.

3. Workloads: Benchmarks evaluated bulk data transfers (1 MB to 100 MB single-stream transfers) and multi-object web browsing workloads (simulating HTTP/2 object downloads over multiplexed streams).

B. MultipathTester (MNM 2019) Mobile Testbed Setup:

To capture dynamic wireless effects absent in Mininet, De Coninck and Bonaventure developed MultipathTester, an Android measurement application deployed on commercial smartphones operating over live commercial LTE and Wi-Fi networks ***[Published Research: MultipathTester 2019, Section 2]***. The methodology evaluated handover latency and transport throughput during physical user mobility across urban cellular and Wi-Fi networks.

C. Limitations of Emulation Testbeds and Environmental Blockers:

Emulated environments (Mininet/NetEm) isolate protocol logic but introduce significant experimental artifacts ***[Our Analysis]***:

1. Middlebox Absence: Mininet virtual Ethernet (veth) links operate over clean kernel namespaces, completely bypassing the middlebox behaviors (TCP option stripping, segment coalescing, UDP throttling) that govern real-world Internet paths.

2. Synthetic Loss vs. Burst Fading: NetEm uniform random loss fails to emulate wireless burst loss profiles characterized by Gilbert-Elliott Markov models.

3. Host Resource Contention: Running sender, receiver, and network emulation on a single shared CPU introduces scheduling jitter and distorts high-throughput user-space UDP processing.

Environmental Notice: As noted in Section II, the team attempted to deploy a local Mininet testbed, but encountered stock WSL2 kernel omissions (lacking CONFIG_MPTCP) and Multipass virtualization driver failures, leading to the deliberate

focus on rigorous literature analysis ***[Our Analysis]***.

VIII. Evaluation

Synthesis of Published Empirical Findings (No Invented Data):

We synthesize published empirical measurements from peer-reviewed literature, examining how MPTCP and MP-QUIC behave across diverse path conditions ***[Our Analysis]***:

A. Throughput Aggregation under Symmetric Paths:

Under clean, symmetric path conditions (identical bandwidth and RTT, 0% packet loss), published evaluations demonstrate that both MPTCP and MP-QUIC achieve near-ideal throughput aggregation, achieving 90% to 96% of the theoretical aggregate capacity (Bandwidth_1 + Bandwidth_2) ***[Published Research: CoNEXT 2017, Section 4.1; Paasch 2014]***. In symmetric environments, the default Lowest-RTT-First (MinRTT) packet scheduler fills both subflow pipes in proportion to their bandwidth-delay products, minimizing out-of-order packet arrival at the receiver.

B. The Asymmetric RTT Penalty and Receive-Window Collapse:

When paths exhibit significant delay divergence (e.g., Path 1: 10 ms RTT, Path 2: 100 ms RTT), empirical studies reveal a severe performance penalty ***[Published Research: CoNEXT 2017; Paasch 2014]***. Under standard MinRTT scheduling, packets dispatched over the slow path arrive 90 ms after concurrent packets dispatched over the fast path. In MPTCP, the receiver cannot deliver out-of-order bytes to the application due to the single connection-level Data Sequence Number (DSN). Consequently, fast-path packets accumulate in the receiver socket buffer, awaiting missing slow-path bytes. Once the connection-level receive window fills, the receiver advertises a zero-window, stalling transmission across the fast path entirely. In severe asymmetry scenarios, published results demonstrate that dual-path MPTCP throughput collapses to less than 50% of the throughput achieved by the fast path operating alone ***[Published Research: CoNEXT 2017; Paasch 2014]***.

C. Head-of-Line Blocking under Loss Asymmetry:

When one path suffers packet loss (e.g., Path 1: 0% loss, Path 2: 2% loss), MPTCP's connection-level byte-stream abstraction enforces global Head-of-Line (HOL) blocking: a single lost packet on Path 2 stalls delivery of all subsequent data across all subflows until subflow-level TCP retransmission succeeds ***[Our Analysis]***. Conversely, published benchmarks of MP-QUIC demonstrate dramatic resilience during multi-stream transfers: because QUIC isolates byte-stream offsets per stream, packet loss on Path 2 stalls only the specific stream whose frame was lost; all other streams continue uninterrupted delivery to the application ***[Published Research: CoNEXT 2017, Section 4.2]***. However, under single-stream transfers, MP-QUIC experiences connection-level stalling identical to MPTCP ***[Published Research: CoNEXT 2017]***.

D. Handover and Failover Dynamics:

Published measurements over commercial cellular and Wi-Fi networks indicate that MP-QUIC achieves significantly faster failover recovery than MPTCP during abrupt link disconnections ***[Published Research: MultipathTester 2019]***. When an active Wi-Fi link drops, an MPTCP subflow must undergo repeated, exponentially backed-off Retransmission Timeouts (RTO) before the sender declares subflow failure and reinjects stranded packets onto the cellular link, introducing multi-second stalls. In contrast, MP-QUIC leverages the Probe Timeout (PTO) and explicit PATH_ABANDON signaling to rapidly detect path loss and immediately migrate or reinject in-flight stream frames onto the surviving path ***[Published Research: MultipathTester 2019; Specification: draft-ietf-quic-multipath-21, Section 5.3]***.

E. Synthesis Table of Published Empirical Literature:

The following matrix summarizes empirical findings documented across primary published studies, highlighting testbed setups, key observations, and validity caveats ***[Our Analysis]***:

Table 2: Synthesis of Published Empirical Literature and Benchmarks

Study & Year	Testbed Environment	Protocols Evaluated	Workload Tested	Key Empirical Findings	Validity Caveats
De Coninck & Bonaventure (CoNEXT 2017)	Mininet emulator; tc-netem delay & loss injection	MPQUIC (quic-go fork) vs. MPTCP (Linux v0.96)	1MB–100MB bulk transfers; HTTP/2 web objects	Near-ideal aggregation (95%) under symmetric RTT; MPQUIC achieves 3x higher throughput under 2% loss due to stream isolation.	Evaluated pre-IETF Google QUIC; single shared PN space; clean emulation without middleboxes.
MultipathTester (MNM 2019)	Commercial Android smartphones over live LTE & Wi-Fi	MPQUIC vs. MPTCP (Linux out-of-tree) vs. single-path	Dynamic web page loading & bulk download during mobility	MPQUIC achieves seamless handover without stalls; MPTCP experiences multi-second RTO backoff delays during Wi-Fi drops.	Small geographic sample; cellular data plan billing constraints; device battery thermal throttling.
Paasch et al. (ACM CSWS 2014)	Linux kernel testbed with NetEm path delay asymmetry	MPTCP v0 (out-of-tree kernel fork v0.88–0.89)	Bulk transfers under asymmetric delays (10ms vs. 100ms)	Demonstrated severe receive-window blocking: dual-path throughput drops below single fast-path throughput with MinRTT.	Evaluated legacy MPTCP v0; modern upstream Linux kernel has redesigned receive window auto-tuning.
Internet Adoption Scan (arXiv:2106.07351, 2021)	Global active Internet scanning of top 1M Alexa domains	MPTCP v0 / v1 adoption on public web servers	SYN/ACK option probing & TCP option reflection tests	Less than 0.1% of top web servers support MPTCP; high rate of false positives from middleboxes echoing unknown options.	Scanning restricted to server port 80/443; does not capture closed proxy deployments (e.g., Apple Siri, 3GPP ATSSS).

Table 3: Transport Protocol Selection Recommendations by Application Workload

Application Type	Dominant Quality Metric	Path Utilization Mode	Recommended Protocol	Justification & Failure Modes	Level of Evidence
Adaptive Video (DASH / HLS)	Rebuffering ratio, bitrate stability, startup latency	Backup / Handover mode (or BLEST aggregation)	MP-QUIC (draft-21)	Stream-isolated transfers prevent segment download stalls; MPTCP DSN blocking causes video freeze under cellular loss.	Published Research (CoNEXT 2017; MultipathTester 2019)
Bulk File Transfer	Flow completion time, sustained throughput	Dynamic Aggregation mode (MinRTT / Coupled CC)	MPTCP v1 (RFC 8684) for legacy; MP-QUIC for greenfield	MPTCP leverages kernel zero-copy and NIC hardware offloads (TSO/GRO); MP-QUIC excels under lossy wireless links.	Published Research (CoNEXT 2017; Paasch 2014) & Our Analysis
Real-Time Voice (VoIP)	One-way latency (<150ms), jitter, packet loss	Redundant mode (packet duplication) or Primary+Backup	MP-QUIC with DATAGRAM frames (RFC 9221)	Multipath reordering destroys VoIP jitter buffers; unneeded reliability in TCP induces latency spikes. DATAGRAM avoids retransmission.	Specification (RFC 9221; draft-21) & Our Analysis

IX. Summary

A. Synthesis of Findings:

This study delivered a rigorous, first-principles comparative deconstruction of MPTCP v1 (RFC 8684) and MP-QUIC (draft-ietf-quic-multipath-21). MPTCP v1 achieves backwards compatibility with legacy socket software by operating inside the OS kernel and multiplexing TCP subflows beneath a 64-bit Data Sequence Number space. However, this design incurs severe vulnerabilities to middlebox interference (TCP option stripping, sequence mangling) and suffers from global Head-of-Line blocking when path delays or loss rates diverge. Conversely, MP-QUIC operates in user space over UDP, encrypting transport headers via TLS 1.3, maintaining independent per-path Packet Number spaces, and isolating HOL blocking to individual streams. Nonetheless, MP-QUIC incurs substantial user-space CPU overhead and remains

vulnerable to enterprise firewall UDP throttling ***[Our Analysis]***.

B. Root Causes of Slow Internet-Wide Adoption:

Despite technical maturity, multipath adoption remains constrained by structural Internet barriers ***[Published Research: arXiv:2106.07351; RFC 8041; Our Analysis]***:

1. Middlebox Hostility: Enterprise firewalls and NATs routinely strip unknown TCP options or drop SYN packets carrying unrecognized extensions.

2. UDP Throttling and Blocking: Many corporate networks actively drop or rate-limit UDP traffic on non-DNS ports to prevent DDoS amplification, crippling QUIC.

3. Socket API Deficits: Standard POSIX socket APIs offer no cross-interface abstraction, requiring complex netlink daemons (mptcpd) or custom user-space libraries.

4. Asymmetric Billing and Battery Constraints: Simultaneous cellular and Wi-Fi usage accelerates mobile battery depletion and risks exceeding mobile data caps, prompting operating systems (e.g., Apple iOS) to restrict multipath to backup-only system services.

C. Limitations of This Study:

This investigation is strictly theoretical and literature-based. The team conducted no empirical experiments and deployed no custom software. Furthermore, published literature predominantly reflects legacy codebases (MPTCP v0 out-of-tree Linux, 2017 pre-IETF QUIC), which do not fully capture the optimizations of upstream Linux 6.x or MP-QUIC draft-21 ***[Our Analysis]***.

D. Concrete Directions for Future Work:

1. Empirical Benchmarking of Draft-21: Deploy standardized physical testbeds running picoquic across real 5G and Wi-Fi links once draft-21 is published as an RFC.

2. Linux eBPF Schedulers: Evaluate modern eBPF-based packet schedulers in Linux 6.6+ to evaluate dynamic path-weighting algorithms (BLEST, ECF).

3. 3GPP ATSSS Integration: Investigate hybrid access aggregation proxies combining fixed broadband and cellular networks in telecom environments ***[Our Analysis]***.

IX. Reproducibility

GitHub Repository: https://github.com/Krina-Paghadar/CS331-T007-MPTCP-vs-MPQUIC ***[Our Analysis]***.

Confirmation of Theoretical Scope: In strict adherence to academic honesty and course guidelines, there are no experimental scripts, testbed codebases, or proprietary measurement traces to reproduce. The team authored no code and conducted no empirical runs ***[Our Analysis]***.

Verification and Traceability Guide: Every architectural claim, protocol mechanism, mathematical formulation, and empirical metric cited in this technical report is fully verifiable against primary, authoritative public sources ***[Our Analysis]***:

1. MPTCP v1 Mechanics: Traceable directly to RFC 8684 (Sections 3.1–3.6) and RFC 6356.

2. MP-QUIC Mechanics: Traceable directly to draft-ietf-quic-multipath-21 (Sections 3–6), RFC 9000, and RFC 9002.

3. Linux Upstream Implementation: Traceable to the official Linux kernel documentation (net/mptcp) and the multipath-tcp.org upstream wiki.

4. Empirical Findings: Traceable directly to De Coninck & Bonaventure (CoNEXT 2017, Section 4) and MultipathTester (MNM 2019, Section 3).

5. Study Plan & Cross-Training Drill: The repository contains the complete 323-question foundational curriculum (mptcp-mpquic-understanding-plan.md) and 4-session execution plan (mptcp-mpquic-plan-condensed.md), detailing the rigorous cross-training methodology executed by the team ***[Our Analysis]***.

References

[References do not count toward the 10-page limit.]

[1] A. Ford, C. Raiciu, M. Handley, O. Bonaventure, and C. Paasch, "TCP Extensions for Multipath Operation with Multiple Addresses," RFC 8684, Internet Engineering Task Force (IETF), Mar. 2020.

[2] A. Ford, C. Raiciu, M. Handley, and O. Bonaventure, "TCP Extensions for Multipath Operation with Multiple Addresses," RFC 6824, Internet Engineering Task Force (IETF), Jan. 2013.

[3] C. Raiciu, M. Handley, and D. Wischik, "Coupled Congestion Control for MPTCP," RFC 6356, Internet Engineering Task Force (IETF), Oct. 2011.

[4] A. Ford, C. Raiciu, M. Handley, and S. Barre, "Architectural Guidelines for Multipath TCP Development," RFC 6182, Internet Engineering Task Force (IETF), Mar. 2011.

[5] J. Iyengar and M. Thomson, "QUIC: A UDP-Based Multiplexed and Secure Transport," RFC 9000, Internet Engineering Task Force (IETF), May 2021.

[6] J. Iyengar and I. Swett, "QUIC Loss Detection and Congestion Control," RFC 9002, Internet Engineering Task Force (IETF), May 2021.

[7] Y. Liu, Y. Ma, Q. De Coninck, O. Bonaventure, C. Huitema, and M. Piraux, "Multipath Extension for QUIC," draft-ietf-quic-multipath-21, IETF Working Group Draft, Mar. 2026.

[8] O. Bonaventure, C. Paasch, and G. Froentjes, "Use Cases and Operational Experience with Multipath TCP," RFC 8041, Internet Engineering Task Force (IETF), Feb. 2017.

[9] Q. De Coninck and O. Bonaventure, "Multipath QUIC: Design and Evaluation," in Proc. 13th Int. Conf. on emerging Networking EXperiments and Technologies (CoNEXT '17), Incheon, Republic of Korea, Dec. 2017, pp. 160–166.

[10] Q. De Coninck and O. Bonaventure, "MultipathTester: Comparing MPTCP and MPQUIC in Mobile Environments," in Proc. IEEE/IFIP Network Operations and Management Symposium (NOMS), Taipei, Taiwan, Apr. 2019, pp. 898–903.

[11] M. Honda, Y. Nishida, C. Raiciu, A. Greenhalgh, M. Handley, and H. Tokuda, "Is It Still Possible to Extend TCP?," in Proc. ACM Internet Measurement Conference (IMC '11), Berlin, Germany, Nov. 2011, pp. 181–194.

[12] C. Raiciu, C. Paasch, S. Barre, A. Ford, M. Honda, F. Duchene, O. Bonaventure, and M. Handley, "How Hard Can It Be? Designing and Implementing a Deployable Multipath TCP," in Proc. 9th USENIX Symposium on Networked Systems Design and Implementation (NSDI '12), San Jose, CA, Apr. 2012, pp. 399–412.

[13] C. Paasch, S. Ferlin, O. Alay, and O. Bonaventure, "Experimental evaluation of multipath TCP schedulers," in Proc. ACM CSWS Workshop, Singapore, Nov. 2014, pp. 27–32.

[14] M. Froemmgen et al., "From Single Lane to Highways: Analyzing the Adoption of Multipath TCP in the Internet," arXiv preprint arXiv:2106.07351, June 2021.

Appendix A: Team Work Distribution and Member Rating

[Appendices do not count toward the 10-page limit.]

A.1 Work Distribution

Ratings are agreed by the whole team and must be UNIQUE no two members may receive the same rating (no ties). Scale: 1–10 (10 = highest contribution). With 4 members, use four different values (e.g., 10, 9, 8, 7).

[Note: Member ratings are left blank below for unanimous team agreement and signature prior to final submission.]

#	Member Name	Roll No.	Tasks Done	Rating (1–10, unique)
1	Kaushal Rathi	24110290	QUIC fundamentals (RFC 9000/9002) & MP-QUIC draft-21 analysis; packet number space comparison; presentation lead.	
2	Krina Paghadar	24110169	MPTCP v1 architecture (RFC 8684), subflow management, Linux kernel upstream evolution, and historical adoption review.	
3	Pratik Kumar Dhakar	24110277	Coupled congestion control theory (RFC 6356 LIA, OLIA, BALIA), fairness metrics, emulation methodology, and end-to-end byte tracing.	
4	Swamini More	24110211	Published empirical literature synthesis (CoNEXT 2017, MultipathTester 2019), application QoE mapping, and recommendations.	

Appendix B: AI Usage Details

B.1 Prompts

Include prompts (or a link to the chat logs) for the most significant uses listed in Table.

Relevant prompt queries and response .md files are referenced below and located in the repository AI_Used/ folder.

#	AI Tool	Purpose	Associated Markdown Log File in AI_Used/
1	Antigravity (Gemini 2.5 / 3.x)	Structural drafting of report sections adhering strictly to course template guidelines, formatting verification, and repository setup.	AI_Used/prompt_01_repo_report_structuring.md
2	Antigravity (Gemini 2.5 / 3.x)	First-principles technical clarification on RFC 8684 (DSS mappings, MP_CAPABLE statelessness) and draft-ietf-quic-multipath-21 (per-path packet number spaces).	AI_Used/prompt_02_protocol_mechanics.md
3	Antigravity (Gemini 2.5 / 3.x)	Analysis of coupled congestion control equations (RFC 6356 LIA, OLIA) and packet scheduler trade-offs (MinRTT vs. BLEST).	AI_Used/prompt_03_congestion_control.md
4	Antigravity (Gemini 2.5 / 3.x)	Critical literature review synthesis (CoNEXT 2017 & MultipathTester 2019) and application QoE mapping for video and VoIP.	AI_Used/prompt_04_literature_synthesis.md
5	Antigravity (Gemini 2.5 / 3.x)	Presentation slide content generation, speaker notes drafting, and rigorous viva question preparation.	AI_Used/prompt_05_presentation_and_viva.md